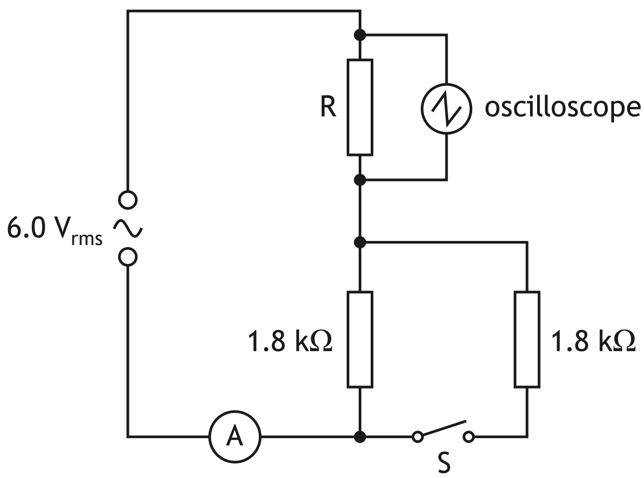
11. A student sets up a circuit using an AC supply of negligible internal resistance as shown.



(a) Switch S is open.

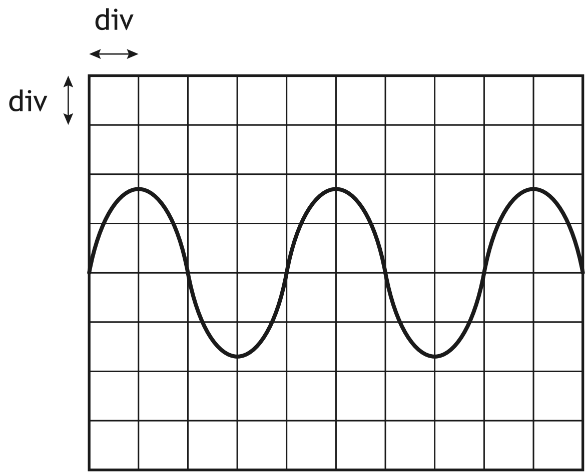
The ammeter displays the rms current. The reading on the ammeter is 2.0 mA.

Determine the resistance of resistor R. **4**

Space for working and answer

11. (continued)

(b) The oscilloscope screen is shown.



The timebase setting on the oscilloscope is 5.0 ms/div.

Show that the frequency of the AC supply is 50 Hz. **3**

Space for working and answer

(c) Switch S is now closed. The settings on the AC supply and the oscilloscope are unchanged.

State whether the amplitude of the trace on the oscilloscope increases, stays the same, or decreases.

You must justify your answer. **3**

[Turn over